



FACULTY OF SCIENCE AND TECHNOLOGY

LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

PROGRAMME: BSSE/BSCS/BIT

YEAR/SEM: YEAR 2 SEM 1

COURSE CODE:BIT2115

PROBLEM NAME: JAVA VARIABLE TYPES

STUDENT DETAILS:

|                 |                  |
|-----------------|------------------|
| NAME OF STUDENT | DURETI EDAO OBSE |
| REG NUMBER      | 25/1707/BSCS-S   |

FACULTY OF SCIENCE AND TECHNOLOGY

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Section 1

Logic Building & Programming

Lines of code & explanations

| Line Number | Explanation                                                                |
|-------------|----------------------------------------------------------------------------|
| 1           | Declares a public class named Course                                       |
| 2           | Opens the course class body                                                |
| 3           | Declares an instance variables courseName                                  |
| 4           | Declares an instance variables numberOfStudents                            |
| 5           | Declares a static variable totalStudents and initializes it to 0           |
| 7           | Declares constructors courseName and numberOfStudents                      |
| 8           | Opens the constructor body                                                 |
| 9           | Assigns incoming parameter to instance field courseName                    |
| 10          | Assigns incoming parameter to instance field numberOfStudents              |
| 12          | Updates the current courses to the shared static                           |
| 13          | Closes the constructor body                                                |
| 14          | Declares an instance method with no return value to display course metrics |
| 15          | Opens the showStatistics method body                                       |
| 16          | Declares and initializes a local variable for pass rate.                   |
| 18          | Prints the string Course to the console                                    |
| 19          | Prints Number of students to the console                                   |
| 20          | Prints Pass percentage to the console                                      |
| 21          | Opens the showStatistics method body                                       |
| 22          | Declares the application execution entry point method.                     |
| 23          | Opens the beginning of the main method                                     |
| 24-28       | Initializes a new course object for the course with number of students     |
| 30-32       | Calls showStatistics on each course object, printing its specific details  |
| 34          | Prints the static cumulative student total                                 |
| 35          | Close the main method body                                                 |
| 36          | Close the Course class body                                                |



FACULTY OF SCIENCE AND TECHNOLOGY

## LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

### Detailed Screenshots and caption descriptions

```
1 public class Course
2 {
3     String courseName; 2 usages
4     int numberOfStudents; 2 usages
5     static int totalStudents = 0; 2 usages
6
7     Course(String courseName, int numberOfStudents) 3 usages
8     {
9         this.courseName = courseName;
10        this.numberOfStudents = numberOfStudents;
11
12        totalStudents += numberOfStudents;
13    }
14    void showStatistics() 3 usages
15    {
16        double passPercentage = 85.0;
17
18        System.out.println("Course: " + courseName);
19        System.out.println("Number of students: " + numberOfStudents);
20        System.out.println("Pass percentage: " + passPercentage + "%");
21    }
22    public static void main(String[] args)
23    {
24        Course java = new Course( courseName: "Java", numberOfStudents: 20);
25
26        Course database = new Course( courseName: "Database", numberOfStudents: 30);
27
28        Course networking = new Course( courseName: "Networking", numberOfStudents: 30);
29
30        java.showStatistics();
```

The screenshot shows an IDE window titled "Course.java" with the following code:

```
1 public class Course
2 {
3     String courseName; 2 usages
4     int numberOfStudents; 2 usages
5     static int totalStudents = 0; 2 usages
6
7     Course(String courseName, int numberOfStudents) 3 usages
8     {
9         this.courseName = courseName;
10        this.numberOfStudents = numberOfStudents;
11
12        totalStudents += numberOfStudents;
13    }
14    void showStatistics() 3 usages
15    {
16        double passPercentage = 85.0;
17
18        System.out.println("Course: " + courseName);
19        System.out.println("Number of students: " + numberOfStudents);
20        System.out.println("Pass percentage: " + passPercentage + "%");
21    }
22    public static void main(String[] args)
23    {
24        Course java = new Course( courseName: "Java", numberOfStudents: 20);
25
26        Course database = new Course( courseName: "Database", numberOfStudents: 30);
27
28        Course networking = new Course( courseName: "Networking", numberOfStudents: 30);
29
30        java.showStatistics();
```

The IDE interface includes a sidebar with project files, a top toolbar with various icons, and a bottom status bar showing the current file path, time, and system information.



FACULTY OF SCIENCE AND TECHNOLOGY

## LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

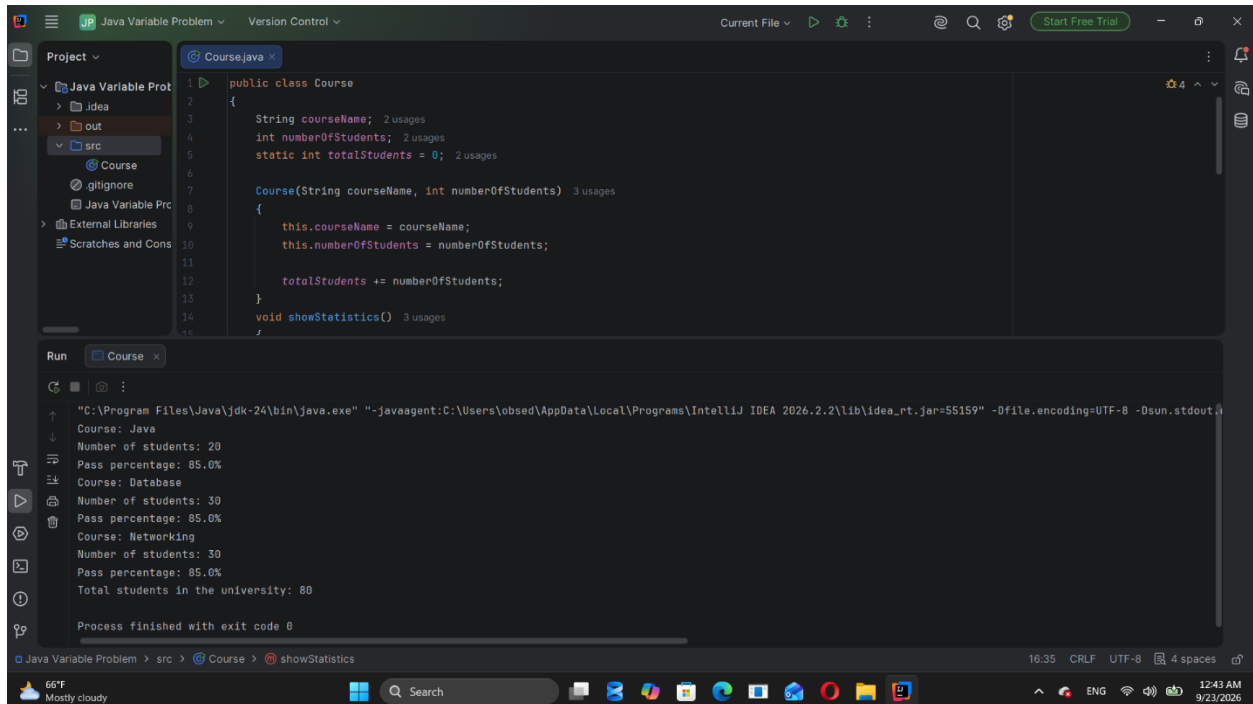
A screenshot of an IDE (IntelliJ IDEA) showing a Java project named 'Java Variable Problem'. The project structure is visible in the left sidebar, showing a 'src' directory containing a 'Course' class. The main editor window displays the 'Course.java' file, which contains the following code:

```
1 public class Course
14 void showStatistics() 3 usages
16 double passPercentage = 85.0;
17
18 System.out.println("Course: " + courseName);
19 System.out.println("Number of students: " + numberOfStudents);
20 System.out.println("Pass percentage: " + passPercentage + "%");
21 }
22 public static void main(String[] args)
23 {
24     Course java = new Course(courseName: "Java", numberOfStudents: 20);
25
26     Course database = new Course(courseName: "Database", numberOfStudents: 30);
27
28     Course networking = new Course(courseName: "Networking", numberOfStudents: 30);
29
30     java.showStatistics();
31     database.showStatistics();
32     networking.showStatistics();
33
34     System.out.println("Total students in the university: "
35         + Course.totalStudents());
36 }
37
38 }
```

The bottom status bar shows the current file is 'Course.java' and the method being executed is 'showStatistics'. The system tray at the bottom indicates the time is 12:45 AM on 9/23/2026.

Figure 1:Class Main showing project structure

### Output and caption descriptions



The screenshot displays the IntelliJ IDEA IDE interface. The top toolbar shows the 'Run' button (a green play icon). The left sidebar contains the 'Project' view, showing a directory structure with 'src' and 'Course'. The main editor window shows the 'Course.java' file with the following code:

```
1 public class Course
2 {
3     String courseName; 2 usages
4     int numberOfStudents; 2 usages
5     static int totalStudents = 0; 2 usages
6
7     Course(String courseName, int numberOfStudents) 3 usages
8     {
9         this.courseName = courseName;
10        this.numberOfStudents = numberOfStudents;
11
12        totalStudents += numberOfStudents;
13    }
14    void showStatistics() 3 usages
15 }
```

Below the editor, the 'Run' console shows the output of the program:

```
Course: Java
Number of students: 20
Pass percentage: 85.0%
Course: Database
Number of students: 30
Pass percentage: 85.0%
Course: Networking
Number of students: 30
Pass percentage: 85.0%
Total students in the university: 80
Process finished with exit code 0
```

The bottom status bar indicates the file encoding is UTF-8 and the current line is 16:35.

Figure 2: Output after running the program

### Output Explanation

From the output provided above, when the program is executed, it will display the output as seen above.



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LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

## Section 2

### Version Control

#### Repository Overview

|                 |  |
|-----------------|--|
| Repository Name |  |
| Repository URL  |  |
| Branches        |  |